# Ahmed Hannan

**Full-Stack Software Engineer | Web, Mobile (iOS/Android) & Systems**

ahmedhanan4@gmail.com • +92 333 149 5251 • linkedin.com/in/ahmedhannan2 • github.com/ahmed-hannan • Islamabad, Pakistan

## Professional Summary

Full-Stack Software Engineer with proven industry experience architecting, deploying, and maintaining high-concurrency web and mobile applications (iOS/Android). Proficient in modern full-stack frameworks (React, Next.js, Node.js, Django, Flutter), parallel systems programming (C/C++, OpenMP, MPI, CUDA), and cloud database management (PostgreSQL, MongoDB, BigQuery). Recipient of a professional Performance Certificate at EmpowerBits for high-impact production engineering under active concurrent user load. Dedicated to writing clean, maintainable, high-throughput code.

## Technical Skills

**Languages:** JavaScript, TypeScript, Python, Dart, C/C++, SQL, HTML5/CSS3, Bash

**Frontend & Mobile:** React.js, Next.js, Flutter (iOS & Android), Tailwind CSS, Redux, State Management, Vite

**Backend & APIs:** Node.js, Express.js, Django, FastAPI, Flask, RESTful APIs, Microservices, WebSockets

**Databases & Cloud:** PostgreSQL, MongoDB, Firebase, BigQuery, Redis, Google Cloud Functions, Cloud SQL

**Systems & Parallel Computing:** Parallel Computing (OpenMP, MPI, CUDA), Multithreading, Concurrency, Socket Programming, Linux

**DevOps & Software Tools:** Git/GitHub, Docker, CI/CD Pipelines, Jira, Agile/Scrum, Selenium, Apache Airflow

## Professional Experience

**EmpowerBits**, Full-Stack Developer — 2026/06 – Present

- Engineered and deployed production-ready web and mobile applications (iOS & Android) actively serving concurrent user traffic.
- Built modular, reactive UI architectures with Flutter and React, optimizing state management to reduce client-side re-renders and latency.
- Designed resilient backend REST APIs and microservices, implementing database query indexing and connection pooling in PostgreSQL.
- Awarded Certificate of Excellence / Performance Certificate for outstanding technical execution, engineering ownership, and rapid feature delivery.

**RedCoast**, Full-Stack Software Engineer — 2025/06 – 2026/06

- Spearheaded full-stack feature implementation and performance optimization across multi-tier client web and mobile applications.
- Managed one-to-one client interactions to translate operational specifications into scalable, production-ready architectural designs.
- Automated integration and regression testing using Selenium and Jira, significantly improving engineering velocity and release stability.
- Conducted large-scale query optimization and analytics on user engagement telemetry via Google BigQuery.

**National Center of AI (NCAI, NUST)**, AI Intern — 2024/08 – 2024/09

- Investigated deep learning architectures with a focus on computational efficiency, resource profiling, and technical documentation.

**FAST – NUCES**, Teaching Assistant (Programming Fundamentals) — 2024/06 – 2024/09

- Assisted in delivering core programming coursework; mentored 50+ students in memory management, dynamic allocation, and C/C++ debugging.

## Featured Software Engineering Projects

**Pose2Play — Real-Time VR & Computer Vision Platform** (Final Year Project) — GitHub: Pose2Play

- Engineered a full-stack interactive rehabilitation platform integrating low-latency computer vision pipelines with a Unity VR simulation engine.
- Developed real-time inter-process communication (IPC) between Python kinematic estimators and Unity C# VR rendering components.

**CRAG: Full-Stack Advanced RAG Platform & Web Dashboard** — GitHub: crag-advanced-rag-pipelines

- Developed a full-stack web platform implementing 7 advanced retrieval strategies (RAG Fusion, HyDE, CRAG, Graph RAG, RRR) on the Meta CRAG corpus.
- Built a Flask REST API backend integrated with FAISS vector indexing and an interactive React + Vite frontend dashboard.

**Parallel Implementation of ParButterfly on Large Bipartite Graphs** — HPC Graph Analytics

- Designed and implemented a scalable parallel algorithm for butterfly-motif counting in billion-edge bipartite graphs using OpenMP and MPI.
- Optimized cache locality, memory bandwidth, and load balancing across multi-core symmetric multiprocessing (SMP) and cluster environments.

**GoMatch & Full-Stack Application Platforms** — GitHub: GoMatch, QuizVerse

- Architected a mobile ride-hailing app replicating Uber/Careem workflows with driver-rider matching, live geolocation tracking, and trip histories.
- Engineered responsive full-stack platforms featuring role-based access control, Google OAuth/JWT authentication, and real-time state management.

## Education

**National University of Computer & Emerging Sciences (FAST-NUCES)**, BS in Computer Science — Graduation: June 2026

- Relevant Coursework: Distributed Systems, Advanced Operating Systems, High-Performance Computing, Database Systems, Software Engineering, Networks, Algorithms.